

SHARVANI CHELUMALLA

☎ 763-568-5982 ✉ ch.sharvani.29@gmail.com 💼 [sharvanichelumalla](#) 🌐 [Sharvani1291](#) 🌐 [Portfolio](#) 📍 Dallas, TX

Data Engineer with **7+ years** of experience building scalable data pipelines, cloud-native data platforms, ETL/ELT workflows, and analytics-ready datasets using **Python, SQL, PySpark, AWS, Azure, Databricks, Snowflake, and Apache Airflow**. Experienced in data lake/lakehouse architectures, data modeling, data quality, orchestration, CI/CD automation, and distributed processing systems supporting reporting, BI, and AI-enabled analytics. Strong collaborator with analysts, engineers, and business stakeholders to translate requirements into reliable, scalable, and production-ready data solutions.

Experience

Data Engineer – Data Modernization

Jul. 2026 – Present

IBM

Dallas, Texas

- Supporting an enterprise **data modernization** initiative on **Azure Databricks**, working across Bronze and Silver lakehouse datasets containing **15.7M+ records** and **1.9M+ telecom orders**.
- Developed **Databricks SQL reconciliation and gap-analysis workflows** across legacy and modernized data pipelines to identify missing orders, ingestion gaps, and source-to-target inconsistencies.
- Performed large-scale **order and milestone validation** across **1.6M+ orders** and **7.6M+ milestone records**, validating complex telecom order lifecycles across multiple enterprise service domains.
- Developed **complex Databricks SQL pipelines** using nested CTEs, window functions, anti-joins, correlated subqueries, and temporal logic to reconcile large-scale order and milestone data across Bronze and Silver layers.
- Validated **order lifecycle and provisioning workflows** across legacy and modernized platforms, tracing submission, provisioning, milestone progression, and completion.
- Built and tested a **Databricks Lakeview API validation dashboard** covering **6 APIs, 466 test batches, 2,800+ orders, and 7.7M+ field-level comparisons**, identifying Databricks-to-external-system JSON mismatches and supporting migration UAT.
- Automated **data quality and exception analysis** for large-scale Databricks datasets, enabling faster identification of anomalous records, reconciliation failures, and migration-impacting data issues.

Data & Application Migration Engineer

Sep. 2025 – Jul. 2026

Yoonefi Technology Solutions LLC

Dallas, Texas

- Led migration of a region-specific nonprofit unit onto the **stacX Platform**, transitioning legacy workloads to a cloud-native architecture spanning **12 source systems** and **28 data pipelines**.
- Re-architected **ETL/ELT workflows** using **AWS Glue, Apache Spark, and cloud storage services** to modernize ingestion, transformation, and migration layers.
- Built a reusable **JSON-driven AWS Glue/Spark mapping framework** supporting schema and data-type casting, default-value handling, and timestamp sanitization across heterogeneous source tables.
- Implemented **incremental and streaming ingestion pipelines** with centralized monitoring and logging, reducing migration cutover downtime from **6 hours to 2 hours**.
- Implemented **ID remapping, de-duplication, and PostgreSQL staging upserts** to preserve referential integrity across accounts, orders, shipping, and related transactional datasets.
- Developed custom parsers for **malformed CSV files and PHP-serialized payloads**, converting irregular source data into normalized JSON structures for downstream processing.
- Defined automated **data quality, source-to-target reconciliation, and validation checks** to ensure migration accuracy and support compliance and audit requirements.
- Optimized migration pipelines to reduce runtime by **35%** and cloud costs by **22%**, while improving pipeline SLA to **99.6%**.
- Developed migration **runbooks, validation checkpoints, and release controls** to enable repeatable onboarding of additional regions onto the stacX Platform.

Data Engineer Co-op
Resolution Life

Aug. 2024 – May 2025
Atlanta, Georgia

- Built an **event-driven data lake ingestion framework** using **AWS Glue, PySpark, S3, Lake Formation, and IAM** to process structured and unstructured data for enterprise analytics and reporting workloads.
- Designed scalable batch and real-time **ETL/ELT pipelines** and reusable **PySpark/Glue templates**, increasing availability of analytics-ready datasets by **35%** while reducing manual processing and source-onboarding effort.
- Enhanced data quality and query performance by **30%** through automated validation, standardized **Glue Data Catalog** metadata and partitioning, and optimization of **Apache Iceberg** tables.
- Supported **Microsoft OneLake integration** and enterprise data lake modernization to centralize storage, improve dataset discoverability, and streamline downstream analytics access.
- Implemented **CI/CD pipelines** using **AWS CodePipeline and Jenkins** for data workloads deployed on **Amazon EKS**, reducing release cycles by **25%**; implemented monitoring and alerting using **CloudWatch and SNS**.
- Prepared curated datasets for **Power BI** by validating transformations and aligning data models with business reporting requirements; also supported **AWS Bedrock**-based AI-enabled insurance data workflows.
- Documented end-to-end **data lake architecture, lineage, operational workflows, and runbooks** in Confluence to improve incident response, technical onboarding, and cross-team knowledge sharing.

Data Engineering Research Assistant
UGA School of Computing (Concurrent with M.S. in CS)

Aug. 2023 – May 2025
Athens, Georgia

- Designed and implemented a fault-tolerant distributed data processing and federated learning framework using transient cloud infrastructure and asynchronous communication, reducing infrastructure costs by **40%**.
- Built scalable distributed workloads using **Kubernetes, Celery, Redis**, and asynchronous task queues across heterogeneous compute environments.
- Integrated **gRPC-based distributed communication** and synchronization workflows, achieving a **7x speedup** in distributed processing and model synchronization.
- Deployed distributed compute workloads using **Azure Virtual Machines (Spot VMs)** and stored training artifacts in **Azure Data Lake Storage (ADLS)** for scalable cloud-native data management.
- Implemented workflow orchestration, monitoring, telemetry, and recovery mechanisms using **Azure Service Bus, Azure Monitor**, and Redis-based coordination.
- Designed scalable **checkpoint management**, metadata synchronization, model state tracking, and workload recovery workflows to improve fault tolerance and reproducibility.
- Built monitoring pipelines to track node availability, workload health, aggregation latency, resource utilization, and distributed job execution metrics.

Associate Data Engineer
Cognizant Technology Solutions

May. 2020 – Jul. 2023
Hyderabad, India

- Built event-driven **data ingestion frameworks** using **AWS Glue, PySpark, Lambda, S3**, and **Redshift** for enterprise analytics and reporting systems.
- Designed scalable **ETL workflows**, cloud-native data lake architectures, and reusable transformation components for large-scale structured and semi-structured datasets.
- Built and managed **AWS data lake** solutions using **S3, DynamoDB, Redshift, Athena, Step Functions**, and **EventBridge** for batch and near real-time processing.
- Supported cloud-based **data migration** and transformation initiatives by developing **Python/SQL** validation scripts, transformation logic, and source-to-target reconciliation workflows.
- Implemented **AWS Lake Formation** governance policies and fine-grained access controls to support secure enterprise data platforms and analytics environments.
- Worked with **Databricks, PySpark**, and scalable transformation frameworks for distributed analytics processing across cloud environments.
- Implemented **data quality validation**, schema enforcement, reconciliation, and operational monitoring workflows to improve reliability across production pipelines.

- Developed CI/CD workflows using **Git**, **Jenkins**, **CloudFormation**, and Bitbucket DevOps, improving deployment consistency and reducing manual errors.

Data Engineering Intern
Research Centre Imarat (RCI) – DRDO

Aug. 2019 – Apr. 2020
Hyderabad, India

- Worked with **engineering and test datasets** generated from electronics and communication systems, using **Python and SQL** for data preparation, analysis, and validation.
- Processed and organized **sensor, test, and system-generated data** to support engineering analysis, experimental evaluation, and technical reporting.
- Developed **Python scripts** to clean, transform, and validate structured engineering datasets, identifying inconsistent, incomplete, and missing measurements.
- Supported **data analysis and validation workflows** for electronics and communication engineering systems by evaluating system-generated measurements and experimental results.
- Collaborated with engineers to document test results, organize technical datasets, and prepare validated data for **engineering reporting and analysis**.

Technical Skills

Programming Languages: Python, SQL, Scala

Data Engineering: Apache Spark, PySpark, SparkSQL, Kafka, Iceberg, Hive, Databricks, Apache Airflow, Snowflake, ETL/ELT Pipelines, Data Modeling, Analytics Engineering

Cloud Platforms: AWS, Microsoft Azure, Microsoft Fabric

AWS Services: AWS Glue, Lambda, EMR, Redshift, Athena, DynamoDB, S3, IAM, Lake Formation, CloudWatch, SNS, EventBridge, RDS

Azure Services: Azure Data Factory, Azure Synapse Analytics, Azure Data Lake Storage (ADLS), Azure Virtual Machines, Azure Monitor, Azure Service Bus

Databases & Storage: Snowflake, PostgreSQL, MySQL, MongoDB

DevOps & Infrastructure: Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, CloudFormation

Backend & APIs: REST APIs, gRPC

Machine Learning & Libraries: PyTorch, TensorFlow, Pandas, NumPy

Visualization: Power BI, Tableau

Education

University of Georgia
Master of Science in Computer Science

Aug. 2023 – May 2025
Athens, Georgia

Vardhaman College of Engineering
Bachelor of Technology in Electronics and Communication Engineering

Aug. 2016 – May 2020
Hyderabad, India

Certifications

Databricks Certified Data Engineer Professional | [Link](#)

Apr 2026

Microsoft Certified: Fabric Data Engineer Associate | [Link](#)

Feb 2026

AWS Certified Solutions Architect – Associate | [Link](#)

Jan 2024

Docker Foundations Professional Certificate | [Link](#)

Mar 2026

SQL (Advanced) | [Link](#)

Mar 2026